First Name
David

Last Name
Naranjo

Email
d.f.naranjohernandez@tudelft.nl

Event Title
SET - Seismology for the Energy Transition

Event Location
Delft University of Technology, Netherlands

Event Date
2027-03-02

Names and emails of event organizers who are current SSA members

David Naranjo — d.f.naranjohernandez@tudelft.nl — January 1st, 2024 (davidn182@hotmail.com)

Describe the meeting's purpose and objective

The purpose of this meeting is to teach state-of-the-art methods and bring together seismologists working on geological storage and geothermal energy, with the objective of advancing the science of microseismic monitoring.

Across Europe, geothermal doublets and geological storage sites, including for CO_2 and underground hydrogen storage (UHS), pose seismologists the same problem despite being commissioned at pace and for different purposes. They involve the injection and, in some cases, withdrawal of fluids from a reservoir, and can perturb stress on pre-existing faults through pressure and temperature changes. Often the only early-warning signal is microseismicity.

What microseismicity means for these operations differs. Microseismicity migrating away from an injection well may trace a cooling reservoir, fluid migration beyond the intended storage complex (such as leakage), or a fault that was already close to failure.

Understanding the origin of microseismicity relies on a measurement problem: detecting events in noisy seismic records, estimating their locations precisely enough to attribute events to a process, and separating operational from background seismicity. Addressing this requires three main ingredients: i) knowledge of the baseline seismicity of the field, ii) an accurate P- and S-wave velocity model for localizing events, and iii) good coverage of the seismic network for detecting and locating the microseismicity.

Two paragraph summary of the event

This three-day meeting at TU Delft trains early-career seismologists to monitor geothermal and geological storage operations. This includes CO_2 storage and underground hydrogen storage (UHS). Day one covers earthquake catalogs. Using the open induced-seismicity Groningen catalog, participants estimate the magnitude of completeness, track the b-value over space and time, and test the criteria that separate induced from natural earthquakes. Day two covers seismic-network design, a method most seismology courses don't include. Participants first learn the principles behind detecting and locating microseismicity, then explore optimal methods for placing sensors. Day three covers seismic imaging. Participants build seismic-velocity models and get hands-on training in active-source data acquisition and processing.

Each day combines a keynote talk, contributed talks, and a hands-on workshop using open data. Participants also visit real infrastructure. On campus, they see TU Delft's geothermal doublet, built for research. They also visit a commercial geothermal doublet in Kwintsheul that supplies heat to local greenhouses. This site provides the data for the practical exercises: S-wave vibroseis and passive-seismometer recordings. The event will also provide an icebreaker and opportunities for collaborations.

Identify the target audience and expected number of attendees

Early-career seismologists, professionals at national agencies and companies working on the monitoring of induced seismicity in geothermal, geological storage, including CCS and UHS, and related subsurface energy operations.

We expect 20 to 30 attendees. Attendance will be free for all participants. If we secure additional sponsorship for travel grants, we would like to sponsor attendees from countries with new renewable energy initiatives.

Describe outreach efforts to those under-represented in seismology and plans for creating an inclusive environment at the meeting

We specifically target two groups: early-career researchers and researchers from countries where seismic monitoring programs for geological storage, including CCS and UHS, and geothermal energy are still developing. By bringing these groups together, we aim to foster an inclusive environment for sharing ideas, knowledge, and innovation.

We will promote the event openly through SSA, EGU, EAGE, and ORFEUS/EPOS, and attendance will be free of charge. To ensure equal access, all workshops will use openly available data, so participants will not need institutional access to complete the exercises. We will publish all workshop materials and code after the meeting. We will also publish a code of conduct in advance and aim for gender and geographic balance among keynote speakers.

PhD students and postdocs will organize the meeting for their peers, creating a welcoming and inclusive environment for the next generation of seismologists.

Please provide a tentative meeting schedule and program

The purpose of this meeting is to build the specific technical capacity that seismic monitoring demands. Three objectives, one per day:

Day 1. Earthquake-catalog construction and statistics. Groningen provides the reference case: two decades of extraction-induced seismicity, an open KNMI catalog, and the most thoroughly analyzed induced sequence in Europe. Participants will work directly with it, estimating magnitude of completeness, tracking b-value in space and time, and applying the rate and spatial criteria used to attribute events to operations rather than tectonics.

Day 2. Seismic-network design. Where sensors are placed determines which microearthquakes can be detected and how well they can be located, yet network geometry is almost always ignored. Optimising it requires understanding what governs those two targets in the first place, so the day introduces the principles of detection and location, signal-to-noise thresholds, azimuthal coverage, and depth resolution, before turning to optimal network designs.

Day 3. Seismic imaging. Accurate hypocentres and focal mechanisms depend on velocity models. We will provide tools for velocity model construction and imaging of the subsurface. We will introduce active-source seismic measurements, including a novel extraction of S-wave velocities using vibroseis that has been used recently at TU Delft, but also passive-source seismic interferometry. We will also visit the DAP-well geothermal operation on campus at TU Delft, and then the geothermal doublet in Kwintsheul, Netherlands.

List the names of your leadership team

David Naranjo — Postdoc (TU Delft), induced-seismicity monitoring, network design, seismic processing — Lead organiser; programme design; SSA contact.

Lada Dvornik — PhD candidate (TU Delft) — Day 1 lead: network design programme and workshop.

Ali Reza Bagheri — PhD candidate (TU Delft) — Day 3 lead: imaging programme and active-source training. Outreach, advertising, and travel-grant selection.

Sjaak van Meulebrouck — PhD candidate (TU Delft) — Logistics, venue, and site-visit coordination.

Sverre Hassing – PhD candidate (TU Delft) - Responsible for finding and coordinating funding.

Eric Verschuur – Associate Professor (TU Delft) - Responsible for finding keynote speakers.

Deyan Draganov — Associate Professor (TU Delft) — Faculty endorser and institutional contact for funds.

Please provide a description of the planning process

PhD students and postdocs organize this event. They already work on geothermal and geological-storage monitoring, including CO_2 storage and UHS, at TU Delft. Each day has one named lead in charge of its keynote, session, and workshop materials. The team meets monthly.

We will confirm keynote speakers and secure the venue, computer lab, and both site visits by September 2026. We will open the call for abstracts in October. We will close abstracts and announce travel grants in December. Two weeks before the meeting, we will send workshop materials and datasets to registrants, so hands-on sessions start productively.

Materials for the catalog-construction day already exist in draft. We are adapting them to the open Groningen dataset. We will prepare and dry-run the other two days internally beforehand.

What is the event's location?

The event takes place at TU Delft's Faculty of Civil Engineering and Geosciences. The faculty provides a lecture room for talks, a space for the poster session, and a computer lab for the hands-on workshops. Participants bring their own laptops and install the required software in advance.

On Day 2, participants visit DAPwell, TU Delft's own geothermal doublet. It is a research facility with borehole seismic monitoring, located on campus within walking distance. This means the Day 2 visit needs no travel arrangements or added cost.

On Day 3, participants visit a commercial geothermal operation in Kwintsheul, 30 to 45 minutes by road from the faculty. This operation heats local greenhouses. It has also recorded low-magnitude seismicity, which is directly relevant to the training.

Few universities can offer both a research doublet and a commercial operation within an hour of the lecture room.

Identify similar meetings and explain how this meeting will be unique

Most courses skip two of the three skills we teach: designing a seismic network and building S-wave velocity models. Most seismologists treat their network as given, so they never learn to design one. And while many programmes build P-wave velocity models, few also build S-wave models, even though S-wave models decide how well we can locate an event and resolve its focal mechanism. The third skill, building and reading an earthquake catalog, rounds out the three-day course, one skill per day. Participants practice every skill on real data, and visit a geothermal doublet on campus and a working commercial operation nearby.

No other event teaches all of this as a hands-on course. The closest comparison is the Schatzalp Workshop on Induced Seismicity. The Swiss Seismological Service runs it, most recently in Davos in March 2025. But Schatzalp is a conference, not a course: people go there to present results, not to learn new skills. Induced seismicity also gets sessions at the SSA Annual Meeting, EGU, and AGU. Seismology training schools do cover detection and location, but none teach network design or velocity models.

How will this event contribute to the global seismological community and advance earthquake science?

This event trains people for a gap the field hasn't filled. Geothermal and geological-storage projects, including CO_2 storage and UHS, are expanding faster than the workforce trained to monitor them. Every new site needs professionals who can design a monitoring network, detect microseismic events, and locate them precisely enough to separate operational seismicity from background activity. Safe use of the subsurface depends on getting this right.

This event treats network design and velocity-model construction as core competencies, not specialist interests. This raises the baseline quality of induced-seismicity catalogs. Better catalogs mean better

hazard assessments for these operations. This event leaves a public legacy beyond the people it trains. Participants leave with working code and documented workflows. We publish everything openly, so the training outlives the event.

This event connects communities that rarely meet: geothermal, CO_2 storage, and UHS monitoring groups. Methods built for one often work for the other. The event also builds a cohort of early-career researchers. They form connections across European and international monitoring programs, at exactly the career stage when such links matter most.

Please include a detailed budget request

This event requests $5,000 in total, covering five items: early-career travel support, Day 3 transport, catering, materials, and rooms and equipment.

Early-career travel support provides partial travel grants for two to three participants, prioritized by need and targeted at smaller or newer monitoring communities. This costs $1,500.

Day 3 transport covers a chartered bus to and from the geothermal doublet in Kwintsheul. This costs $700.

Catering covers coffee breaks and lunch across all three days. This costs $1,500.

Materials cover workshop handouts, poster boards, and unforseens. This costs $300.

Rooms, equipment, and the icebreaker reception cost $1,000.

Describe the specific intended use of SSA funds for this event

SSA funds make this event free for participants, so early-career seismologists can attend and train regardless of their institution's budget. The $5,000 covers everything needed to run the training: $1,500 for partial travel grants that prioritise participants from smaller or newer monitoring communities, $700 for a chartered bus to the geothermal doublet in Kwintsheul on Day 3, $1,500 for coffee breaks and lunch across the three days, $300 for workshop handouts, poster boards, the programme, and SSA sponsorship signage, and $1,000 for rooms, equipment, and the icebreaker reception. Together, these costs let us focus entirely on making a free trainning program for students and early career scientists.

What other sources of funding will support this event?

TU Delft will provide the facilities and rooms for the meeting. We will also seek additional funding from geothermal, CO_2 storage, and UHS companies.

EBN, TNO, KNMI, and SodM can contribute data and keynote speakers. We will also ask them for additional funding, if possible.

Any additional funding we secure will go toward travel support, helping participants from institutes that cannot easily afford the trip.

If the SSA grant is not approved, or not approved in full, how would that impact the event?

Without SSA's support, this event would need to find funding elsewhere. We will reach out to other sponsors, but none can replace the role SSA plays in bringing together early-career researchers, especially those from countries that most need this training and can least afford to travel for it. If we cannot secure enough funding, we would need to charge participants a fee, and we would likely lose exactly the participants this training is meant to reach.